MIL-C-18438B(SHIPS)
14 June 1962
SUPERSEDING
MIL-C-18438A(SHIPS)
3 February 1959

MILITARY SPECIFICATION

CHARGERS, BATTERY, GENERATOR TYPE

1. Scope

1.1 This specification covers generator type battery chargers and associated control equipment for shipboard (marine) use in two wire direct current distribution systems.

1.2 Classification.-

1.2.1 Types.- The battery chargers shall be of the following types as specified (see 6.1):

Type I - Direct current generator and control unit.
Type II - Three phase alternating current (a.c.) generator, rectifier and control unit. (Power rectifiers air blast cooled by means of the generator fan or blower).

1.2.2 Classes.- The battery chargers shall be of the types and classes shown in table I, as specified (see 6.1):

Table I - Classes

Class	Type	Rated watts	Rated volts
1	I	Under 250	7.2
2	I	Under 250	14.3
3	I	Under 250	28.5
4	I	250	14.3
5	I	250	28.5
6	I	500	14.3
7	I	500	28.5
8	I	1000	14.3
9	I	1000	28.5
10	I	2000	28.5
11	I	5000	28.5
12	II	500	14.3
13	II	500	28.5
14	II	1000	14.3
15	II	1000	28.5
16	II	2000	28.5
17	II	5000	28.5

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular, (General Purpose).

MILITARY

MIL-E-1 - Electron Tubes and Crystal Rectifiers.
MIL-R-22 - Resistors, Variable (Wirebound Power Type).
MIL-C-25 - Capacitors, Fixed, Paper - Dielectric, Direct Current (Hermetically Sealed in Metallic Cases) General Specification for.
MIL-C-62 - Capacitors, Fixed, Electrolytic (DC, Aluminum Dry Electrolytic, Polarized) General Specification for.
MIL-A-907 - Antiseize Compound, High Temperature.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-B-3743 - Brushes, Electrical Contact; and Carbon Stock, Electrical Contact, Brushes.
MIL-C-5541 - Chemical Films for Aluminum and Aluminum Alloys.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-Q-9858 - Quality Control System Requirements.
MIL-C-11693 - Capacitors, Feed Through, Radio Interference Reduction, Paper Dielectric, AC and DC (Hermetically Sealed in Metallic Cases).
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-F-15733 - Filters, Radio Interference, General Specification for.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.

FSC 6130

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MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper-Oxide, and Magnesium-Copper Sulfide Types, Naval Shipboard Use.

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-195 - Marking of Connections for Electrical Assemblies.

MIL-STD-242 - Electronic Component Parts (Selected Standards for).

MIL-STD-408 - Brushes, Electrical Contact, Rectangular, Standard Designs and Dimensions for.

MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.

DRAWINGS

BUREAU OF SHIPS

RE 49A501 - Couplings, Spherical Seat, Design Data.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Definitions.-

3.1.1 Minimum full load speed is the minimum generator speed at which the battery charger (hot) will deliver rated load at a voltage which is constant to within plus or minus 3 percent of rated value.

3.1.2 Maximum full load speed is the maximum generator speed at which the battery charger (hot) will deliver rated load at a voltage which is constant to within plus or minus 3 percent of rated value.

3.1.3 The battery charger shall be defined as "hot" when its operating temperature is stabilized by operation at the lowest speed at which rated current can be obtained and voltage can be maintained constant within plus or minus 3 percent of rated value.

3.1.4 Full load speed range is the range of generator speed at which the battery charger (hot) can deliver rated current and maintain voltage to within plus or minus 3 percent of rated value.

3.1.5 Full or rated load is defined as the current delivered by the battery charger when operating under a load corresponding to the rated watts and volts specified for the particular type and class of battery charger shown under 1.2.2.

3.1.6 Cut in speed is defined as the speed where sufficient generator voltage is produced to energize and close the control unit relay which connects the battery charger to the battery.

3.2 Detail requirements.-

3.2.1 Material.- Material shall be new, and unused, and shall be first class in every respect. Further, such material shall insure continued operation under the conditions of heat, moisture and vibration specified herein. Military standard parts shall be used whenever they are suitable for the purpose. When military standard parts are not available, AN parts shall be used. Where military standard and AN parts are not suitable, the use of commercial grade basic materials equivalent to the minimum characteristics and performances of the applicable military specifications is encouraged. When military standard parts, AN parts or parts referenced in military specifications are suitable and commerical parts are used, drawings shall show the commercial part and the corresponding military standard, AN or military specification part which can be used as a replacement for the commercial part. Source inspection of basic material shall be kept to a minimum consistent with quality control as required to assure that construction and performance requirements have been met. Materials, specifically prohibited by the requirements of MIL-E-917 shall not be used. Further, the use of magnesium and its alloys is prohibited.

3.2.2 Interchangeability.- All similar parts, including repair parts, furnished under the specification and manufactured to the same drawings shall be strictly interchangeable without the necessity of further machining or hand fitting.

3.2.3 The design and construction of the equipment shall be such as to obtain conformance with the requirements herein and with good engineering practice. Workmanship shall be first class in every respect.

3.2.4 Unless otherwise specified in the contract or order (see 6.1), all equipment shall be of the two wire (ungrounded) design.

3.2.5 Insulation material, insulation varnish, varnish treatment, clearance and creepage distances shall meet the requirements of MIL-E-917.

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3.2.6 Battery chargers covered by this specification shall be considered acceptable provided the particular design, type or model meets the material, construction and performance requirements of this specification and has demonstrated its ability to meet al least one of the following conditions:

(a) The design has been previously accepted and has been successfully evaluated as a shipboard installation.
(b) The design has been successfully applied in commercial marine service or has been laboratory tested for a period of at least 1000 hours. During the first 1000 hours in service or during the first 1000 hours of laboratory tests, the battery charger shall have operated without the necessity of replacing, repairing or adjusting of any part.
(c) Successful completion of the 1000 hour test as specified (see 4.5.19) prior to delivery of production units.

3.2.6.1 If condition (3.2.6(a)) can be met, a technical dissertation shall be provided with bid data. This dissertation shall include:

(a) Available shop drawings showing assembly and installation feature of the battery charger.
(b) Name and address of the activity which had previously procured the proposed battery charger.
(c) Identification of contract or purchase orders for previous procurements of the battery charger.
(d) Manufacturer's type or model number for the battery charger.
(e) Identification of ships, boats or craft on which proposed charger has been installed.

3.2.6.2 If condition 3.2.6(b) can be met, a technical dissertation shall be submitted with bid data. This dissertation shall include the information required by 3.2.6.1(a), 3.2.6.1(d) and the following:

(a) At least two names and addresses of commercial marine activities using the proposed battery charger or the name and address of the laboratory which conducted evaluation tests.

3.2.7 Battery chargers shall be designed and constructed for operation when connected to a battery in any state of charge from discharge to full charge, at any speed below the zero load speed point and up to maximum full load speed and at any ambient temperature between minus 30°F. and 122°F.

3.2.8 Generators (including power rectifiers) shall be designed for operation when rotated up to 180 degrees about the axis of the generator shaft and when permanently inclined up to 30 degrees in any direction from the normal horizontal position of the generator shaft. Control units shall be designed for operation when permanently inclined up to 30 degrees (in any direction) from the normal mounting position.

3.2.9 Use of aluminum.- The use of aluminum or aluminum alloys (except aluminum material designed to carry current) shall conform to the following:

3.2.9.1 Unpainted surfaces of aluminum or aluminum alloys shall be treated to conform to the requirements of MIL-A-8625. Surfaces to be painted shall be treated to conform with the requirements of MIL-C-5541.

3.2.9.2 Contact between dissimilar metals shall be avoided as much as possible. Where the use of dissimilar metals is unavoidable, the aluminum alloys shall have their faying surfaces anodized.

3.2.9.3 Where threaded surfaces or faying surfaces may require disassembly in service, an antiseize compound conforming to MIL-A-907 shall be used. For the assembly of aluminum alloy parts, through bolting is preferred using cadmium or zinc plated steel bolts and nuts; this applies particularly to comparatively highly stressed assemblies or those subject to vibrating loads. Where the use of tap bolts or machine screws is necessary, these shall be threaded into steel inserts in the aluminum alloy. The steel inserts shall be secured in such a manner as to prevent their turning or backing out. No alloys of copper, such as brass or bronze, will be acceptable for use in threaded contact with aluminum alloys.

3.2.10 All hardware such as springs, bolts, nuts and studs shall be made of corrosion resistant material or shall be treated for corrosion resistance.

3.2.11 Parallel operation.- Unless otherwise specified in the contract or order (see 6.1) class 1 through class 7, class 12 and class 13 shall not be designed for parallel operation. Class 8 through class 11 and class 14 through class 17 shall be designed so that they can operate as single units and in parallel. When generators are operating in parallel, each generator shall not deliver more than 100 percent of its rated load continuously or 110 percent rated load intermittently at any speed from "cut in" to maximum full load speed and at any total load from zero to a load equal to the sum of the ratings of machines in parallel. There shall be no periodic pulsation of loads between generators; the voltage regulation shall conform with the requirements herein; and

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there shall be no maloperation of control units components (such as chattering of load relay or cut-out relay contacts) at any speed or load noted above.

3.2.12 Vibration.- Normally, generators and power rectifiers will be mounted on and driven by the prime mover. Control units will normally be bulkhead mounted adjacent to the prime mover. Accordingly, generators and rectifiers mounted within the generator enclosure shall meet the performance requirements specified herein under the mechanical shock and stresses resulting from the internally excited and torsional vibration characteristics of the prime mover. In addition, the equipment shall be designed to operate satisfactorily under the environmental vibration conditions onboard boats and craft. These conditions involve vibration frequencies which vary from zero to approximately 1500 cycles per minute with amplitudes of vibration as much as plus or minus 0.030 inch. Control units shall be designed to operate satisfactorily under the specified environmental vibration conditions.

3.2.13 Enclosures.- Except as otherwise specified (see 6.1), generator enclosures for class 1 through class 9, and class 12 through class 15 battery chargers shall be totally enclosed so that when water is sprayed on the outside of the enclosure, water will not enter the enclosure and cause maloperation or damage to components. Unless otherwise specified, generator enclosures for class 10, 11, 16 and 17 shall be of the dripproof protected type which shall be constructed so that drops of water falling vertically shall not enter the equipment and cause damage to components when the generator is rotated 180 degrees and when the generator, rectifier and control unit are inclined as specified in 3.2.8. Control units enclosures shall be of the totally enclosed type so that water sprayed on the outside of the enclosure will not enter the enclosure and cause maloperation or damage to components.

3.2.14 Maximum temperature rises for various parts (based on 122°F ambient) shall be as specified in table II.

3.2.15 Radio interference reduction.-

3.2.15.1 All battery chargers shall be designed to produce minimum conducted and radiated radio interference within the frequency range of 150 kilocycles and 20 megacycles without use of radio interference suppression filters or capacitors. The following design features shall be considered to reduce radio interference:

(a) Largest practical number of armature coils and commutator bars.
(b) Shallow armature slots.
(c) Short pitch armature windings.
(d) Minimum eccentricity of commutators and collector rings.
(e) Balanced design to eliminate shaft currents without the use of auxiliary brushes.

3.2.15.2 When the contract or order (see 6.1) requires battery chargers to meet the requirements of MIL-I-16910, adequate shielding of components shall be provided and spherical seat couplings, as shown on drawing RE49A501, shall be provided to permit use of shielded type cables by the installing activity. These couplings shall be complete with matching fittings for securely attaching cable shielding to the housing. Where a military standard battery charging system is specified, the number and size of couplings shall be as specified for the standard. Where a military standard is not available, the number and size of couplings shall be as specified in the contract or order (see 6.1). In general, the number and size of couplings shall be based on the size of shielding material used to enclose cables between the: (a) generator and the control unit; (b) control unit and the enclosures for radio interference reduction filters and capacitors where these filters and capacitors are not mounted within the enclosure of the control unit or generator. When radio interference reduction is accomplished by means of filters and capacitors, filters shall meet the requirements of MIL-F-15733 and capacitors shall conform to MIL-C-11693. Where these filters and capacitors are mounted outside the enclosures of the generator and control unit and where required to meet radio interference limits, these filters and capacitors shall be installed within an enclosure with appropriate spherical seat connectors. Where radio interference reduction can be accomplished by use of feed thru capacitors, these capacitors shall be installed in the control unit. Feed thru capacitors shall conform to MIL-C-11693 and shall be of appropriate capacitance, voltage and current rating. In addition, these feed thru capacitors shall be capable of withstanding (without damage), an insulation resistance test voltage of 500 volts direct current for a period of two minutes. If the proper rating of feed thru capacitor is not available, the contractor shall submit a proposed capacitor to the bureau or agency concerned for approval prior to assembly of production units. The feed thru capacitor arrangement for radio interference reduction is preferred over the scheme which requires separate mounting of filters and capacitors. Feed thru capacitors shall be furnished with each battery charger. Unless otherwise specified in the contract or order (see 6.1), filters and capacitors requiring separate mounting shall not be furnished with battery chargers. However, details of these capacitors and filters, and the enclosure in which they are installed, shall be shown on the drawings required in 3.3.

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Table II - Maximum temperature rise °C

Item	Method of measuring temperature	For speeds at or below 140 percent of minimum full load speed	For speeds above 140 percent of minimum full load speed
Contacts			
(a) Solid	Method 1 [1/]	70	70
(b) Solid, silver fayed	Method 1	80	80
Bolted terminals and terminal studs			
(a) Not plated	Method 1	50	50
(b) Silver plated	Method 1	65	65
Bearings	Method 1	60	60
Rheostats - and - resistors			
(a) Bare resistor material	Method 1	375	375
(b) Resistor embedding material	Method 1	300	300
Capacitors-Air within 1/2 inch of case	Method 1	35	35
Commutators and collector rings			
(a) Class A insulation	Method 1	75	65
(b) Class B insulation	Method 1	85	75
(c) Class F Insulation	Method 1	110	100
(d) Class H Insulation	Method 1	145	135
Coils (Single layer)			
(a) Class A insulation	Method 1	60	50
	Method 2 [2/]	70	60
(b) Class B insulation	Method 1	80	70
	Method 2	70	80
(c) Class F insulation	Method 1	105	95
	Method 2	115	105
(d) Class H insulation	Method 1	130	125
	Method 2	150	145
Coils (multi-layer)			
(a) class A insulation	Method 1	55	45
	Method 2	65	55
(b) class B insulation	Method 1	75	65
	Method 2	85	75
(c) class F insulation	Method 1	100	90
	Method 2	110	100

See footnotes at end of table.

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Table II - Maximum temperature rise °C (Cont'd.)

Item	Method of measuring temperature	For speeds at or below 140 percent of minimum full load speed	For speeds above 140 percent of minimum full load speed
Coils (multi-layer) (Cont'd)			
(d) class H insulation	Method 1 Method 2	125 145	120 140
Cores, poles and other mechanical parts in contact with insulation	(Same as for insulation which the parts contact)		

Semiconductor Devices - Permissible temperature rise of semiconductor devices is dependent on operating conditions. Maximum permissible case temperatures shall be shown on equipment drawings.

1/ Method 1 involves measurements with thermometer or thermocouple.
2/ Method 2 involves measurements by resistance method.

3.2.16 Application of battery charger.- The following requirement is the responsibility of the activity designing belt drive systems for battery chargers furnished in accordance with this specification. In applying the battery chargers specified herein, the belt drive and pulley arrangement should be designed to provide at least 1500 hours of reliable belt operation (at least 1500 hours of operation before replacement of belts is necessary) with the particular prime mover involved and under the operating conditions (heat, moisture and vibration) specified herein. Further when the prime mover speed varies such as in propulsion applications, every effort should be made to design a belt drive system which will permit maximum use of the battery charger full load speed range so that battery charging power can be delivered over most or all of the prime mover speed range without causing maloperation of control components such as chattering of load relay or cut-out relay contacts. Where the prime mover operates at constant speed, the belt drive arrangement shall be designed so that the generator is driven at a speed at least 200 r.p.m. above the minimum full load speed.

3.2.17 Generators.-

3.2.17.1 The generators furnished under this specification are intended for belt drive from small boat gasoline, diesel or turbine propulsion and auxiliary generators set prime movers. A shaft extension (with keyway or splined shaft extension (as specified, see 6.1), to accommodate driving pulleys shall be provided. The shaft extension shall be of sufficient length to accommodate a two sheave pulley suitable for use with two "B section size V-belts." Holes shall be drilled and tapped into the generator frame to permit installation of mounting brackets. The number, dimensions and location of the mounting holes and details of the shaft extension for installation of driving pulleys shall be as specified in the contract or order (see 6.1). Unless otherwise specified in the contract or order (see 6.1), belts, pulleys, mounting bases and mounting brackets shall not be provided with generators.

3.2.17.2 The direction of rotation for Type I direct current generators shall be as specified on the contract or order (see 6.1). The direct current generator identification plate shall include information to indicate the direction of generator rotation, or a separate identification plate shall be provided to show the direction of rotation. Generators for Type II battery chargers shall be designed for both directions of rotation without modification of any type.

3.2.17.3 The generator shall be constructed to permit ready inspection of brushes, commutators or collector rings without removing end brackets or the fan. If the brush rigging assembly on direct current generators is not a fixed position type, the brush yoke shall be doweled to the end bracket or other positive means shall be provided to prevent the brush yoke from shifting from the neutral position.

3.2.17.4 Brush holders.-

3.2.17.4.1 General requirements.- Contact surfaces shall not react chemically to produce corrosion and sticking of brushes when brushes or graphite, electro-graphite, or copper-graphite are used. The brush holders shall be of such design

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that the brushes will not bind in the brush holders. The brush-feeding device shall ensure proper brush contact under all normal conditions of brush and commutator or collector ring wear.

3.2.17.4.2 Brush stop.- Means shall be provided to prevent brushes from being displaced to such a position that they may bind in their brush holders and not return to their normal operating position. The devices used shall not unduly complicate replacement of the brushes, require the use of a special design of brushes, or result in cumbersome or intricate design of brush holders.

3.2.17.4.3 Pressure finger.- The pressure finger shall be of such design that when operating with a brush having a smooth top surface (not notched) or a hammer clip, it will not slide off the brush or clip under any operating conditions. When the brush is omitted from the holder, a stop shall prevent the finger from making contact with the commutator or collector ring.

3.2.17.4.4 Brush holder spring.- Springs shall be of either the adjustable type or constant pressure type. Where adjustable springs are used means shall be provided for adjusting the brush pressure without special tools.

3.2.17.4.5 Types.- Brush holders shall be either full box type or open front type. The open front type shall be of such design that the frame work covers not less than 50 percent of the front side.

3.2.17.4.6 Terminal screw.- A separate terminal screw shall be provided for securing the brush shunt terminal lugs to the brush holder.

3.2.17.5 Brushes shall be of a grade, temperature rating and type as specified in MIL-B-3743. The brush design and dimensions shall be as specified in MIL-STD-408.

3.2.17.6 The generator armature and fan shall be keyed to the shaft or securely fastened to the shaft by other means approved by the bureau or agency concerned to prevent movement of the armature and fan with respect to the shaft.

3.2.17.7 Pre-tinned banding wire of corrosion resistant steel shall be used on the armature end turns or other banding material may be used if specifically approved by the bureau or agency concerned.

3.2.17.8 A rabbet fit shall be provided between the frame and end brackets. This fit shall be adequate to maintain alignment of components and, where class 1 through class 9 and class 12 through class 15 battery chargers are involved, this fit shall be adequate to prevent entry of water. Gaskets shall not be used between the generator end brackets and the frame. However, use of non-hardening gasket compound in conjunction with a rabbet fit will be permitted provided the compound does not interfere with the fit of parts.

3.2.17.9 Totally sealed prelubricated bearings are preferred over the type which requires lubricant to be added periodically. Where totally sealed bearings are used they shall be type 120 of FF-B-171. Where bearings require lubricant to be added they shall be type III, class 1 of FF-B-171. Bearings shall be adequately lubricated before the generator is shipped from the manufacturer's plant.

3.2.17.10 Compression type grease cups shall be used on all generators where the type of bearing requires grease to be added periodically. Accessible grease drains and plugs shall be provided at the bottom of the bearing housings.

3.2.17.11 Collector rings and commutators.-

3.2.17.11.1 Collector rings and commutators shall run true, have a polished surface, and shall be free from porosity and hard spots. The collector rings or commutator with associated retaining flange or device shall be securely attached to the generator shaft to prevent axial or rotational movement of the rings or commutator with respect to the generator shaft.

3.2.17.11.2 The commutator shall have the insulation between bars undercut by removing all insulation to form a groove not exceeding 3/64 of an inch. The wearing depth of commutators shall be not less than 1/8 inch in diameter for commutators up to 2 inches in diameter and not less than 3/16 inch for commutators over 2 inches in diameter. Commutators shall be cured at higher than operating temperatures and shall be of such construction throughout as to insure that they will hold their shape in service and obviate the necessity of frequent refinishing to eliminate high bars or flat spots. After assembly of the commutator, all crevices or joints at each end (such crevices and joints are usually located between the bars and retaining flange) shall be completely filled with appropriate material and sealed over with insulating varnish. In addition, the creepage paths between the bare copper segments and the bare metal retaining flanges at each end of the commutator shall be given a heavy coat of insulating varnish.

3.2.17.11.3 Collector rings shall have a wearing depth not less than 3/32 inch in diameter for rings up to 2 inches in diameter, and 1/8 inch in diameter for rings over 2 inches in diameter.

3.2.17.12 The generator shaft shall be protected against corrosion by a coating of insulating varnish on

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all portions of the shaft except those portions (bearing and pulley seats, threaded sections and similar portions) where clearances or fits must be maintained between the shaft and other parts of the generator. This varnish coating may be applied during varnish treatment of the armature windings.

3.2.17.13 The dimensions, contour and speed-load characteristics shall be as specified in the contract or order (see 6.1). Where a military standard is specified, the dimensions, contour and speed-load characteristics shall be as required for the military standard.

3.2.17.14 Dynamic balance.- The generator armature shall be dynamically balanced so that the remaining unbalance in either plane or correction does not exceed 1/10 ounce-inches.

3.2.17.15 Speeds.- Unless otherwise specified in the contract or order, the maximum permissible operating speeds for direct current generators shall be 6000 r.p.m. and the maximum operating speeds for alternating current generators shall be 8000 r.p.m.

3.2.17.16 The alternating current generator shall be of the stationary armature type. Unless otherwise specified in the contract or order, initial excitation power for generators shall be initiated by residual magnetism features in the generator.

3.2.17.17 Terminal leads, boards and boxes.- All terminal leads shall be flexible stranded cable or wire. An insulating sleeve or other suitable means shall be provided to prevent abrasion of lead insulation. Leads shall be secured in a reliable manner to prevent their coming in contact with moving parts or being chafed by contact with stationary parts. Leads shall terminate in studs which shall be secured to terminal boards in a manner to prevent turning of the studs when tightening or loosening the nuts used to secure leads or lead connectors. Unless otherwise specified in the contract or order (see 6.1), terminal boxes shall be provided and shall be securely bolted or welded to the frame. The boxes shall be totally enclosed and provided with an entrance plate or plates suitable for drilling and installation of cable, or installation of the spherical seat couplings required by 3.2.15.2. The terminal board shall be securely mounted within the terminal box in such a manner as to permit convenient connection of external leads to terminal studs. All terminal studs shall be identified by appropriate permanent markings on studs or the terminal boards. Wherever practicable, markings shall be in accordance with MIL-STD-195.

3.2.18 Rectifiers.-

3.2.18.1 Rectifiers shall be either silicon or selenium types. When both types are suited to an application, the preference shall be silicon over selenium.

3.2.18.2 Silicon rectifiers.- Individual silicon rectifiers shall be selected from Appendix I. In addition, they shall comply with the requirements of the individual specification sheets and maximum rating and rerating values specified in table I of Appendix I. Nonlisted silicon rectifiers shall require specific approval as required by MIL-E-917 for nonpreferred parts. In addition, they shall either be covered by an individual specification sheet or they shall comply with the requirements of an individual specification sheet listed in Appendix I for a similarly constructed device (axial lead, stud-mounted, glass package, and so forth) except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic value contained in the applicable specification sheet. Individual diodes shall be accessible for replacement.

3.2.18.3 Selenium rectifiers.- Selenium rectifiers shall be supplied in accordance with the requirements of MIL-R-18281.

3.2.18.4 Rectifiers used in control circuits shall be mounted within the enclosure of the generator or control unit. Power rectifiers shall be mounted within the enclosure of the generator to permit rectifiers to be air blast cooled by means of generator cooling fan.

3.2.19 Voltage reference diodes.- Voltage reference diodes shall be selected from Appendix I. They shall comply with the maximum rating and rerating requirements specified therein. Nominal 10 watt, 5 to 6 volt regulator diodes having substantially zero temperature coefficient are also permitted provided they are procured to individual sheets of MIL-S-19500 except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in that specification. All other devices will require specific approval as required by MIL-E-917 for nonpreferred parts; in addition, they shall either be covered by an individual specification sheet, or they shall comply with the requirements of an individual specification sheet listed in Appendix I for a similarly constructed device substituting characteristic values peculiar to the devices to be used. Individual diodes shall be accessible for replacement.

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3.2.20 Voltage regulator devices.- Voltage regulator devices shall be supplied in accordance with the requirements of MIL-E-1 and MIL-S-19500 and applicable individual specification sheets thereunder as specified in Appendix I. In addition to listing individual devices from which selection shall be made, Appendix I also lists maximum ratings and derating factors which must be observed in application of these devices. Nonlisted devices will require specific approval as required by MIL-E-917 for nonpreferred parts. In addition, they shall either be covered by an individual specification sheet, or they shall comply with the requirements of an individual specification sheet listed in Appendix I for a similarly constructed device except that the device manufacturer's latest published commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable specification sheet.

3.2.21 Transistors.- Individual transistors shall be selected from Appendix I. In addition, they shall comply with maximum rating and rerating requirements specified in that table. All transistors shall be procured to the requirements of MIL-S-19500 and, where issued, individual military specification sheets under MIL-S-19500. If not specifically covered by such a sheet, they shall comply with the tests and inspection procedures of an individual military specification sheet listed in Appendix I or MIL-STD-701 for a similarly constructed device except that the manufacturer's latest commercial specification values applicable to the device (or approved values based on the commercial product) shall be substituted for the characteristic values contained in the applicable specification sheet. Types not listed in Appendix I will require specific approval as required by MIL-E-917 for nonpreferred parts. The requirements stated herein for preferred types shall apply for all transistors approved for use in this equipment. Individual transistors shall be accessible for replacement.

3.2.22 Rheostats and resistors shall be in accordance with MIL-R-22 or MIL-R-15109.

3.2.23 Except as noted herein, capacitors shall be paper-dielectric fixed capacitors of the extended foil construction conforming to MIL-C-25. Styles and ratings listed in MIL-STD-242 shall be used where possible. For direct current applications, the sum of the highest d.c. voltage level plus the peak a.c. voltage to be applied shall not exceed 70 percent of the working voltage of the capacitor. This 70 percent rerating factor shall be in addition to voltage derating with temperature as required by individual specification sheets under MIL-C-25. The above 70 percent rerating factor for d.c. applications shall be reduced to 60 percent for 60 cycle a.c. applications and to 50 percent for 400 cycle applications. For high values of capacitance or where severe space and weight limitations are imposed, dry-electrolytic aluminum foil fixed capacitors conforming to MIL-C-62 are permitted with the following limitations:

(a) Use only in d.c. circuits.
(b) Do not derate in voltage. From ratings available, select one equal or as nearly as possible to the maximum voltage imposed on the capacitor by the conditions of the application.
(c) Select capacitors of the plug in type and mount them with a retaining device. Styles and ratings shall be as listed in MIL-STD-242 where possible.
(d) Do not apply if ambient temperature at the capacitor can be expected to go above +85° C or below -40°C.

3.2.24 Control units.-

3.2.24.1 Control units shall be designed to accomplish the following:

(a) Automatically connect the battery charger to the battery when the generator reaches "cut in" speed and automatically disconnect the battery charger from the battery when the generator speed drops below the "cut-in" value and a reverse current of not greater than 12 percent of rated current flows from the battery to the charger. In type I equipment, the closing voltage shall be of sufficient magnitude to charge batteries and prevent maloperation of the control unit components. In type II equipment, the closing voltage may be below a value needed to charge batteries where the blocking action of power rectifiers provide adequate protection against reverse current or maloperation of components until the voltage builds up to a value which will charge the batteries. When specifically approved by the bureau or agency concerned, the "cut-out" or load relay may be closed and opened manually on Type II equipment.
(b) Automatically maintain the direct current voltage constant to within plus or minus 3 percent of rated voltage at any load between zero and 95 percent rated load within the full load speed range. This requirement applies after the battery charger has attained normal operating temperatures (hot temperature) as defined herein. Where the battery charger has not attained operating temperature, the voltage shall be maintained within plus or minus 5 percent of rated voltage at any

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load up to 95 percent of rated value at any speed within the full load speed range.

(c) Automatically limit the maximum battery charger output to between 95 and 105 percent of rated current to prevent overloading of the battery charger should it be connected to a discharged battery.

(d) Automatically control the output of battery chargers so that paralleled chargers will meet the parallel operation requirements specified herein.

3.2.24.2 The control unit shall be suitable for bulkhead mounting or mounting on the engine. In general, vertical bulkhead mounting is preferred. If engine mounting is required, the contract or order shall so specify (see 6.1). Vibration isolators shall be supplied with the control unit if they are required to insure satisfactory operation of control unit components.

3.2.24.3 Where a military standard is specified (see 6.1), the weight, dimensions and contour of control units shall be as required for the standard. Where a military standard has not been established or where a military standard cannot be used due to major installation problems, the contract or order (see 6.1) shall specify weight, dimension and contour characteristics.

3.2.24.4 Unless otherwise specified (see 6.1), the requirements of 3.2.17.17 herein apply to terminals, terminal boards and terminal boxes for control units.

3.2.24.5 Voltage regulator and current limiting elements may be static in design (utilizing semiconductor devices) or may be of the vibrating contact or carbon pile types.

3.2.24.6 The voltage regulator element in the control unit shall permit the regulated direct current voltage to be adjusted between 95 and 105 percent of rated voltage in steps not to exceed 0.2 volt. This requirement applies at any load from zero to rated load and at any speed within the full load speed range. Prior to shipment, each control unit shall be set to maintain voltage at rated value.

3.2.24.7 The control unit design shall be such that there will be no chattering (intermittent closing and opening) of "cutout", or load relay contacts when the control unit acts to connect or disconnect the battery to the battery charger.

3.2.25 Identification plates.-

3.2.25.1 A separate plate shall be furnished for each generator and control unit. The plates shall be attached to the part of machinery which will not ordinarily be renewed during its normal service life. These plates shall be located in a readily accessible position where they can be read at all times without danger to personnel. Identification plates shall be made of brass, nickel-copper alloy, corrosion resistant steel or anodized aluminum. The plate markings shall be either etched, stamped, engraved, metal photo or cast in such a manner as to produce permanent and durable markings to last for the anticipated life of the equipment. All etchings, engravings and stampings shall not be less than 0.003 inch deep. The characters on cast plates shall be raised to at least 0.03 inch. All engraved, stamped or direct etched markings shall be filled with black paint, enamel or lacquer.

3.2.25.2 The minimum data to be marked on identification plates shall include the following:

(a) Major unit name (that is: a.c. generator, d.c. generator, control unit, and so forth.
(b) Federal stock number.
(c) Manufacturers type or model number for the generator and control unit.
(d) Manufacturer's name and serial number.
(e) Government contract number.
(f) Blank space for government inspector's stamp.
(g) Maximum permissible ambient temperature.
(h) Voltage and current (or wattage) rating.

3.3 Drawings.-

3.3.1 General requirements.- Drawings shall be in accordance with the requirements of MIL-D-963 and as specified herein.

3.3.2 Kinds of drawings.- Equipment design and installation details shall be shown on Class A drawings conforming to MIL-D-963 and contractual data shall be shown as class B (certification data) drawings conforming to MIL-D-963. Drawings shall be of the following types, as specified in the contract or order (see 6.1).

(a) Preliminary drawings.
(b) Final drawings.

3.3.2.1 Preliminary drawings.- Where approval of drawings is required by the contract or order (see 6.1), preliminary drawings shall be furnished as specified in the contract or order (see 6.1) (except as noted herein). These drawings may be either blue prints, blue line prints or equal. In the event the design has been previously approved for use under this specification and copies of approved class A drawings are already on hand at the bureau or agency concerned, submission of certification data (class B drawings) along with identification data for the class A drawings involved, will be sufficient for drawing approval action purposes.

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3.3.2.2 Final drawings.- The number of sets of final drawings furnished shall be specified in the contract or order (see 6.1). Final drawings shall be either microfilm, tracings or direct reading blackline prints as required by MIL-D-963. Where microfilm is furnished, a microfilm index shall be provided. A proposed index shall be submitted for bureau or agency approval prior to actual microfilming. Drawings on the microfilm shall be arranged as component groups headed by the assembly drawing. Drawings on microfilm shall be preceded by the microfilm index. In the event that copies of (class A) final drawings are on hand at the bureau or agency concerned, a direct reading blackline print of the class B (certification data) drawing will adequately cover final drawing requirements.

3.3.3 Contents of drawings.-

3.3.3.1 Information to be shown on class A drawings shall be as follows:

3.3.3.1.1 Descriptive data.-

(a) Identification data.- Class and type as specified herein (see 1.2.2) and manufacturer's type, class or model number for the generator and control unit.
(b) Reference specification.- A statement as follows: "The battery charger shown hereon conforms to MIL-C-18438 dated (show date) with the following exceptions: (List all exceptions as applicable)."
(c) Performance.-

(1) A battery charging system performance curve to show the load-speed characteristics at rated voltage and load for all generator speeds up to maximum operating speed.
(2) A battery charging system performance curve to show the horse power and torque required to drive the charging system at maximum output and at any speed up to maximum operating speed.

(d) Generator classification.-

(1) Enclosure
(2) Ambient temperature
(3) Mounting
(4) Insulation class
(5) For direct current generators:

a. Armature core diameter, core length and nominal air gap.
b. Number of main poles and number of commutating poles.
c. Commutator diameter, length, wearing depth, undercut, insulation between bars, insulation to ground and creepage and clearance distance to ground.

(6) For alternating current generators:

a. Stator core diameter, core length and nominal air gap.
b. Rotor core diameter and core length.
c. Number of collector rings, ring inside and outside diameters, ring material, wearing depth, insulation to ground and creepage and clearance distance to ground.

(7) Brush information, consisting of manufacturer's grade, military grade, thickness, width, length, top and bottom bevel, number of brush studs and number of brushes per stud.
(8) Shaft, bearings and lubrication:

a. Shaft material
b. Specific type and size of bearing
c. Type of lubrication

(9) Weight of complete generator
(10) Installation details including:

a. Overall and principal dimensions.
b. Location and size (including thread size data) for mounting holes.
c. Details of shaft extension provided for installing the driving pulley.
d. Location and size of the terminal box and terminals.
e. Identification of each terminal in the terminal box.
f. Location and size of spherical seat couplings provided.
g. Location and identification of rectifiers installed within the generator enclosure.

(11) A list of commercial parts used and the corresponding military standard, AN or military specification part which may be used as a replacement for the commercial part.

3.3.3.1.2 Winding data.-

(a) Armature winding data to include type of winding, number of slots, number of coils, number of conductors in parallel, pitch of coil, size of bare conductor, size of conductor over insulation, feet of wire per coil, weight of copper per coil, and resistance of wound armature in ohms at 25°C.

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(b) Armature winding diagram facing connection end.
(c) Cross section view of armature slot showing coils and all insulation.
(d) Cross section view of field poles showing coils and all insulation.
(e) Field coil winding data to indicate turns per coil, size of bare conductor, size of conductor over insulation, number of conductors in parallel, inside and outside dimensions of coil (where coil is form wound) and resistance of wound coil in ohms at 25°C.
(f) Field coil connection diagram.
(g) The following insulation data shall be shown:

(1) Identify all insulation material used by indicating the type, thickness, applicable government specification and where applied.
(2) Sequence of all winding and insulation operations.
(3) Tape application, including type, number of layers, amount of overlap, and so forth.

(h) Ground insulation, slot cell insulation, coil spacers and separators, end turn insulation, insulation under banding material, support pads, distance that slot cell insulation extends beyond laminations, slot wedges, method of bringing out and insulating leads from each coil and method of securing end turns.
(i) Varnish treatment schedules including type of varnish, impregnation methods, time cycles and baking temperatures for wound coils prior to installation as well as treatments for completed subassemblies.

3.3.3.1.3 Construction details.-

(a) An assembly drawing showing a side and end view of the generator and control unit with all parts identified with piece numbers. Separate detail views of subassemblies shall be provided where the assembly drawing does not clearly show the information required.
(b) Method of securing all rotating elements to the generator shaft. This includes bearing inner races, fans, commutators, collector rings, armatures, rotating field assemblies, spiders, spacers and sleeves.
(c) Method of securing all stationary elements to the frame.
(d) Bearing housing construction, fits and tolerances.
(e) Method of bringing cables out of frame into terminal box and method of protecting these cables against chafing at sharp metallic corners.
(f) Method of securing leads to the terminal boards in the terminal box.
(g) Overall dimensions of terminal box and terminal box covers and the number and size of cover bolts.
(h) A detail working drawing of the generator shaft including material characteristics, dimensions, finishes and tolerances.
(i) Sectional views of the generator commutator and collector ring assemblies, showing holders and brush holder rigging (including studs and insulation). The materials of the brush holder body, brush holder spring and brush holder rigging shall be indicated.
(j) Direction of generator rotation and the direction of air flow through or around the generator frame.
(k) Winding data for transformers, reactors and magnetic amplifiers including number of turns, tape, wire size and type, terminal identifications, direct current resistance between terminals at 25°C., nominal voltage between terminals, volt-ampere ratings for transformers, and current ratios for current transformers. Complete material details including applicable specifications for each material used, details of core boxes, layer and coil insulation, ground insulation, method of impregnation, treatment and potting methods shall be included.
(l) Resistor data shall include, type, resistance, voltage and wattage (or current) rating, fixed or adjustable and any other special data which may be necessary for ordering exact duplicates.
(m) Rheostat data shall include, nominal diameter, number of plates, connections between plates, resistance, voltage and current rating and any other special data which may be necessary for ordering exact duplicates.
(n) Capacitor data shall include type, capacitance, voltage rating, standard military type number and part designation. Also indicate the maximum a.c. (peak and rms) and d.c. (average) voltages which will be applied to the capacitor in the application and the condition at which these maximum voltages occur.
(o) Complete winding data for all relay coils, including number of turns, wire size and type, direct current resistance at 25°C, nominal voltage and current between terminals. Complete material details including applicable specifications for each

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material used, details of core boxes, layer and coil insulation, ground insulation, and varnish treatment method shall be included.

3.3.3.1.4 Semiconductor devices.- Each device shall be identified in the list of material by manufacturer's part or drawing number to permit direct ordering Joint Electronics Devices Engineering Council (JEDEC number is adequate). Also, mounting details and application (not device capability) data shall be provided as follows:

(a) Rectifiers using silicone diodes.

(1) Circuit symbols and piece number.
(2) Maximum reverse voltage peak and r.m.s. (complete rectifier circuit).
(3) Maximum load current d.c. amperes (complete circuit).
(4) Maximum diode case temperature at design ambient-use for diodes rated against core temperature.
(5) Maximum ambient temperature directly adjacent to diodes-use for diodes rated against ambient temperature - no cooling plates.
(6) Cooling plate size, thickness and material where so mounted.
(7) Air velocity if forced cooled (linear feet per minute.)
(8) Circuit (for example, single phase bridge, and so forth).
(9) Total diodes in circuit.
(10) Diodes in series per arm.
(11) Diodes in parallel per arm.
(12) Description of insulation hardware if insulated from a cooling plate.

(b) Rectifiers using selenium cells (in stacks or cartridges).

(1) Circuit symbol(s) and piece number.
(2) Maximum a.c. voltage applied to the rectifier circuit.
(3) Maximum d.c. load current through rectifier circuit
(4) Maximum cell temperature (shall not exceed 75°C).
(5) Cell dimensions (3 by 3 inches, 5 by 6 inches, and so forth).
(6) Air velocity if forced cooled (linear feet per minute).
(7) Rectifier circuit (single phase bridge, and so forth).
(8) Total stacks or cartridges used in circuit.
(9) Stack or cartridge circuit (if different from rectifier circuit).
(10) Cells in series per arm of stack or cartridge.
(11) Cells in parallel.
(12) Special requirements specified in the purchase order but not covered in MIL-R-18281 but considered necessary for this application (such as a tighter reverse current limit for magnetic amplifier or control application).

(c) Voltage regulator diodes.

(1) Circuit symbol(s) and piece number.
(2) Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes and still give required equipment performance ___ volts to ___ volts).
(3) Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) plus or minus ___ mv.
(4) Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds) ___ ma.
(5) Maximum transient reverse current under any operating condition ___ ma for ___ sec. duration and ___ frequency of occurrence.
(6) Maximum ambient temperature at diode (no cooling plate).
(7) Maximum case temperature of diode (complete for diodes rated against case temperature).
(8) Cooling plate size, thickness and material where used.
(9) Air velocity if forced cooled (linear feet per minute).
(10) Total diodes in circuit.
(11) Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
(12) Description of insulation hardware if insulated from a cooling plate.

(d) Voltage reference diodes (see regulator diodes also).

(1) Circuit symbol(s) and piece number.
(2) Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s) ___ volts to ___ volts.

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(3) Error permitted in an installed voltage reference diode under all conditions of operation, plus or minus ____ mv.
(4) Circuit design current through voltage reference:

a. Nominal or average value: ___ ma.
b. Regulation about nominal value: plus or minus ____ ma.

(5) Maximum ambient temperature (include internal heat rise above outside design ambient): ____ a.
(6) Number of voltage reference elements in series: ____ (operation in parallel is not permitted).

(e) Transistors.-

(1) Circuit symbol(s) and piece number.
(2) Maximum collector to emitter voltage ____ volts.
(3) Maximum collector to base voltage ____ volts.
(4) Maximum collector current ___ ma.
(5) Maximum base current ___ ma.
(6) Maximum collector power dissipation ___ mw.
(7) Maximum ambient temperature (include internal heat rise at transistor at 50°C design ambient ___ °C.
(8) Maximum core temperature of transistor at 50°C. equipment operation ___ °C. (Where rating is based on case temperature.)
(9) Air velocity of forced air cooled.
(10) Cooling plate size, thickness and material where so mounted.
(11) Description of insulating hardware if insulated from a cooling plate and any other special features which may have been provided.

3.3.3.1.5 Schematic diagram.-

(a) A schematic diagram of the complete system to clearly represent the operation of the equipment. The following features shall be incorporated.

(1) Each part, such as resistors, capacitors, and so forth shall be identified by symbol (R-resistor, T-transformer, L-reactor, and so forth).
(2) In addition to the part designation, the numerical value of all resistances, capacitances and inductances shall be noted adjacent to the part.
(3) Identification of all terminals shall be shown.
(4) Simplicity in reading and understanding the operation of the system shall be stressed throughout.

3.3.3.1.6 Wiring diagram.-

(a) A wiring diagram for each of the major subassemblies and interconnection of the subassemblies. The following features shall be incorporated in the drawing:

(1) All parts shall be shown in correct relation to physical location within a particular enclosure.
(2) Terminals of all assemblies and subassemblies shall be clearly shown and marked.
(3) Designation of parts shall conform to those assigned in the schematic diagram. Also, the piece numbers corresponding to those shown in the list of material shall be indicated.
(4) Wire size, type, applicable specifications, and type of terminals shall be shown in tabular form.
(5) Lead identification.
(6) Continuous lines to represent leads between components need not be drawn if the lead identification used clearly indicates the termination point of each lead.

3.3.3.1.7 Description of theory of operation.

(a) A brief description of theory of operation of the system to enable the approval authority to understand the circuitry.

3.4 Technical manuals - Where approval of manuals is required, preliminary manuals shall be submitted for approval as specified in the contract or order (see 6.1). Final technical manuals shall be furnished in the quantity and shall be distributed as specified in the contract or order (see 6.1). Where technical manuals are specified they shall meet the requirements of Type 1 in MIL-M-15071 and shall include the following information:

(a) Chapter 1 - General information.-

(1) Section 1 - Battery charger data shall include a component list and characteristic data (model, type and rating) of the generator and control unit.
(2) Section 2 - Introduction. A brief explanation of what a battery charger is and what it will do should be included.

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This section should also include a block diagram of the system (showing major components) and a brief explanation to show the function of each component.

(b) Chapter 2 - Detailed description.- A detailed description of major parts and assemblies with sketches, drawings or photographs.

(c) Chapter 3 - Principle of operation.- Schematic diagram of various assemblies (particularly of control unit assemblies) with an explanation of the function of each part.

(d) Chapter 4 - Installation.-

Section 1 - Installation instructions.
Section 2 - Check out procedure before start-up.

(e) Chapter 5 - Operating instructions.-

(1) Section 1 - Safety precautions.
(2) Section 2 - Start up procedure (including adjustments and checks to be made after start up has been completed).
(3) Section 3 - Parallel operation (If battery charger is designed for parallel operation).
(4) Section 4 - Shutdown.

(f) Chapter 6 - Maintenance.-

(1) Section 1 - Preventative maintenance. Routine inspection and care of equipment.
(2) Section 2 - Corrective maintenance.-

a. Trouble shooting charts including tables showing points of measurements of voltage, current, resistance, and so forth.
b. Details of test set ups required including details of all instruments used and schematic diagrams.
c. Step by step procedure for determining component which has failed.
d. Details of all adjustments which should be made after repair has been completed.

(g) Chapter 7 - Parts identification.-

(1) Section 1. Introduction to include brief description and interpretation of parts descriptive data.

(2) Section 2. Parts tabulation.- Photographs or exploded views of various assemblies shall be included to clarify parts tabulation.

(h) Chapter 8 - Drawings, diagrams and curves.- These shall include copies of the drawings, diagrams and curves showing the information required by 3.3.3.1.1 through 3.3.3.1.6 herein.

3.5 Repair parts.-

3.5.1 Onboard repair parts.-The total number of sets of onboard repair parts shall be as specified in the contract or order (see 6.1). Where onboard repair parts are specified, the total number of identical parts furnished in each set of repair parts shall be based on the number of battery chargers per ship or activity as specified in the contract or order (see 6.1) and on table III . Where a repair part is not normally separated from an assembly (such as contacts which cannot be separated from supporting members), then the complete assembly shall be furnished in lieu thereof as a repair part. Where matching of parts is required in the equipment, the matched set of parts shall be considered one part and furnished as a repair part. Except as specified herein, the word "sets" referenced in Table III is defined as the total number of parts installed in one battery charger.

3.5.1.1 Where rectifiers are used, they shall be furnished in a quantity sufficient to replace all rectifying elements connected to a common d.c. terminal of the rectifier circuit in which it is used. Where the rectifier item of supply is used in more than one circuit configuration, the quantity furnished shall satisfy the requirements specified herein for the configuration requiring the largest quantity of the rectifier item of supply. Quantities furnished as specified herein shall be considered one set of rectifiers for the particular size and type involved.

3.5.2 Stock repair parts shall be furnished as specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The contractor shall provide and maintain a quality control system acceptable to the government for the equipment covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858.

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Table III - Quantities of onboard repair parts to be furnished per ship or activity.

Part required	Number of identical battery chargers installed per ship or under maintenance responsibility of Naval overhaul activity.							
	1	2	3	4	5-8	9-20	21-5	51-100
Bearings, sets	1	1	1	1	2	3	4	5
Brush holders, complete with spring	1	1	1	1	2	3	4	5
Brush holder springs	1	2	3	3	5	6	7	8
Brushrigging insulation, sets	1	1	1	1	2	3	4	5
Brushes, sets	2	3	4	4	5	6	8	12
Complete generator	0	0	1	1	1	2	3	4
Capacitors each size and type	1	1	1	1	2	3	4	5
Contacts each size and type	1	1	1	1	2	3	4	5
Resistors each size and type	1	1	1	1	2	3	4	5
Springs each size and type	1	1	1	1	2	3	4	5
Rheostats (each size & type if springs and contacts are not replaceable)	1	1	1	1	2	3	4	5
Rectifiers, (set each size & type) (See 3.5.1)	1	2	3	3	4	6	7	8
Relay armature assemblies (each size and type)	1	1	1	1	2	3	4	5
Switches (each size and type)	1	1	1	1	2	3	4	5
Transistors(each size & type)	1	2	3	3	4	6	7	8
Voltage regulator diodes (each size and type)	1	2	3	3	4	6	7	8
Voltage reference diodes (each size and type)	1	2	3	3	4	6	7	8
Complete control unit	0	0	1	1	1	2	3	4

4.3 Sampling.-

4.3.1 Inspection lot.- All battery chargers of the same type, class, and power rating presented at one time shall be considered a lot for the purposes of acceptance inspection.

4.3.2 Sampling for quality conformance inspection.-

4.3.2.1 Group A tests.- Each battery charger shall be subjected to each of the group A tests specified in 4.4.1.

4.3.2.2 Group B tests.- A random sample of battery chargers shall be selected in accordance with table IV from each inspection lot and subjected to each of the group B tests specified in 4.4.2.

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Table IV - Sampling for group B tests.

Number of machines in inspection lot	Sample size; number to be tested	Number of nonconforming machines found in each sample tested	
		Acceptance number	Rejection number
1	1	0	1
2 to 15	2	0	1
16 to 40	3	0	1
41 to 65	5	0	1
66 to 110	7	0	1
111 to 180	10	1	2
181 to 300	15	2	3
301 to 500	25	3	4
501 to 800	35	5	6

4.3.2.3 Group C tests.- A battery charger of a particular type, class, and power rating shall be selected at random from the first production lot and subjected to the group C tests specified herein. Once every two years thereafter, a machine of the same type, class and power rating shall be selected from a production lot and shall be subjected to these tests. A copy of the latest group C test results with adequate abstract shall be forwarded to the bureau or agency concerned to verify conformance with the group C test requirements herein.

4.4 Quality conformance inspection (to be conducted by the manufacturer at the place of manufacture).-

4.4.1 Group A tests.- Each battery charger shall be subjected to the tests specified herein. Failure to conform with this specification for any group A test shall be counted as a defect and the battery charger shall not be offered for delivery. These battery chargers may be offered again for inspection provided all deficiencies have been corrected by the manufacturer.

Group A tests	Reference paragraph
Assembly	4.5.1
General inspection	4.5.2
Dielectric	4.5.6
Insulation resistance	4.5.7
Commutation (type I units only)	4.5.13
Overspeed	4.5.15
Lubrication	4.5.16
Control devices	4.5.17
Dynamic balance	4.5.18

4.4.2 Group B tests.- Each of the sample machines selected in accordance with 4.3.2.2 shall be subjected to each of the following tests and the results of each test compared with the requirements of this specification. Failure to conform to this specification for any group B test shall be counted as a defect and the machine shall not be offered for delivery. If the number of nonconforming machines in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall not be offered for delivery. These lots may be offered again for inspection provided all machines in the lot have been retested by the manufacturer for the test(s) causing noncompliance and all noncompliance corrected.

Group B tests	Reference paragraph
Resistance of windings	4.5.9
Voltage regulation	4.5.11
Commutation (type II units only)	4.5.13

4.4.3 Group C tests.- The battery charger selected in accordance with 4.3.2.3 shall be subjected to each of the following group C tests and the results compared with this specification. Failure to conform with this specification for any test shall require the manufacturer to correct the noncomplying feature on future production units and on any unit not yet shipped. If investigation reveals the noncomplying feature may exist on units previously accepted and shipped, full particulars concerning defects found, including recommendations for correcting the feature, shall be furnished the bureau or agency concerned.

Group C tests	Reference paragraph
Radio interference[1/]	4.5.3
Enclosure[1/]	4.5.4
Heating	4.5.5
Inclined operation	4.5.8
Resistance	4.5.9
Saturation	4.5.10

See footnote on following page.

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Group C tests	Reference paragraph
Parallel operation	4.5.12
Weight	4.5.14

1/ Radio interference tests shall be conducted only when 3.2.15.2 herein applies. Enclosure tests to be conducted only on totally enclosed generators and control units.

4.4.4 Special tests.- Where 3.2.6(c) applies, a prototype of the battery charger shall be subjected to the tests under 4.5.19.

4.5 Test procedures.-

4.5.1 Assembly.- Tests shall be performed on completely assembled, battery chargers after all components have been adjusted.

4.5.2 General inspection.- Each unit shall be subjected to a thorough examination to ascertain that the workmanship and design conforms to this specification. The fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the normal life service of the unit.

4.5.3 Radio interference.- When the contract or order (see 6.1) requires radio interference to meet the requirements of MIL-I-16910, tests shall be conducted in accordance with MIL-I-16910 for both radiated and conducted interference except as follows:

(a) The generator, control unit and rectifier (if applicable) shall be operated as a complete system without shielding or filtering. Interference readings shall be taken at no load, half load and full rated load with the generator operating at minimum full load speed and with the generator operating at maximum full load speed.

(b) The generator, control unit and rectifier (if applicable) shall be operated as a complete system with shielding and filters installed. Interference readings shall be taken under the load and speed conditions specified herein.

4.5.4 Enclosure tests.- Acceptability of drip-proof protected enclosures shall be determined by visual examination. Acceptability of totally enclosured type enclosures shall be determined by tests which shall consist of spraying the enclosure with water for a period of one hour. During this test each side of the enclosure shall be exposed to water spray for a minimum of 10 minutes. The enclosure shall be considered acceptable if there is no entry of water into the enclosure during this test.

4.5.5 Heating tests.-

4.5.5.1 Conditions.- Heating tests shall be conducted when the battery charger is delivering full load at the minimum and maximum full load speeds.

4.5.5.2 Duration.- Normal load heat runs on battery chargers shall be continued until the steady final temperatures have been attained in all parts. It shall be considered that steady final temperatures have been reached when at least four consecutive readings taken at 15 minute intervals show no change in the temperature of any part of the battery charger.

4.5.5.3 Measurement of the ambient temperature.-

4.5.5.3.1 Unless otherwise specified in the contract or order, battery chargers may be tested at any convenient ambient temperature above 10°C., but whatever the ambient temperature, the maximum permissible temperature rises specified herein shall not be exceeded.

4.5.5.3.2 The ambient temperature thermometers shall be protected from drafts, and from heat radiation, from equipment under test, or from outside sources.

4.5.5.3.3 The value to be adopted for ambient temperature during the tests shall be the mean of the readings of the several thermometers placed as specified in 4.5.5.3.2, taken at four equal intervals of time during the last quarter of the test. During the first part of the heat run, thermometer readings shall be taken at half hour intervals.

4.5.5.3.4 The conditions in the testing room shall be such that the ambient temperature will not vary greatly during the test. A variation of 10°C. or more, during a period of 6 hours or a proportional change for runs of shorter duration shall in no case be exceeded. If the ambient temperature is very irregular during the run or changes rapidly at the end, the test shall be repeated.

4.5.5.4 In determining temperature rise, no correction shall be made for barometric pressure, humidity, or for any deviations of the recorded ambient temperature from the standard ambient temperature of reference.

4.5.5.5 Method of measuring temperature.-

4.5.5.5.1 The method of measuring temperature rise shall be optional with the manufacturer. Only one method of temperature determination shall be required for any particular part. Temperature rise of rotating fields shall be measured by method 2 (see 3.2.14). Collector rings, bearings and mechanical parts shall be measured by method 1 (see 3.2.14).

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4.5.5.5.2 On shut-down, preheated thermometers shall be placed on the armature commutator, core and windings, and banding wire and on all the collector rings. Every precaution shall be taken to reduce to a minimum the period of time elapsing between the stopping of the machine and the application of the thermometers. A curve shall be plotted with temperature readings as ordinates and time as abscissa. That portion of the linear curve starting where successive readings show decreasing temperatures shall be extrapolated back to the instant of shut-down. The temperature at the instant of shut-down as determined in this manner shall be considered the shut-down temperature.

4.5.5.3 Temperature measurements on the generator, and control unit shall be taken to determine conformance with the temperature rise requirements specified in 3.2.14.

4.5.6 Dielectric tests.-

4.5.6.1 The dielectric test shall be made after all other tests have been completed. If the insulation resistance of the winding is known to be lower than specified, because of dirt or moisture or damage to windings, this condition shall be remedied before the application of the dielectric test voltage. Unless otherwise specified, the dielectric test shall be made on the completely assembled unit and not upon individual parts. An exception is made in the case of maintenance parts, such as coils and rotating elements with insulated windings (which require dielectric tests) and feed-thru capacitors used for radio interference reduction.

4.5.6.2 Test voltage.- The dielectric test voltage shall be 750 volts. The frequency of the testing voltage shall be not less than 60 cycles. The voltage wave shall approximate a sine wave. The testing voltage shall be applied continuously for a period of not less than 60 seconds.

4.5.6.3 Measurements of test voltage.- The measurement of the voltage used in the dielectric tests shall be made by means of a voltmeter whereby the meter derives its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer, or through an auxiliary potential transformer.

4.5.6.4 Points of application.- The test voltage shall be successively applied between each electric circuit and the frame with all other electric circuit and metal parts grounded. Interconnected polyphase windings shall be considered as one circuit. The test voltage shall be applied in such a manner as to preclude the possibility of pitting bearings in case of insulation failure.

4.5.7 Insulation resistance (cold).- Insulation resistance test shall be made before the dielectric tests. Prior to the application of the test voltage, the windings of the machine shall be thoroughly discharged. Separate measurements shall be made on the generator armature and field windings. Circuits of equal voltage above ground shall be connected together. Circuits or groups of circuits of different voltage above ground shall be separated. The temperature of the generator windings at the time of the test shall be measured and recorded. Insulation resistance measurements shall be corrected to 25°C. Unless otherwise specified in the contract or order, the insulation resistance test may be conducted at any convenient ambient temperature. The relative humidity at the time of the test shall be measured and recorded.

4.5.8 Inclined operation test.- The battery charger shall be operated in the inclined positions specified in 3.2.8 to determine that operation is satisfactory under these conditions.

4.5.9 Resistance.- The resistance of the generator windings shall be measured and recorded. The temperature at which these measurements were made shall also be recorded.

4.5.10 Saturation data.- No load saturation data (volts versus field amperes) shall be taken with the generator operating at speeds corresponding to minimum and maximum full load speeds.

4.5.11 Voltage regulation test.- The battery charger shall be connected to a battery of applicable voltage rating and to variable resistance which shall be set for approximately 25 percent of full load current at rated generator voltage when measured at the battery charger output terminals. Vary the generator speed from minimum full load to maximum full load speed in increments of approximately 400 r.p.m. full load current. For the 100 percent load run, the resistance load shall be set such that there shall be no operation of the current limiter. In each case record generator current, voltage, speed, and field current. The voltage regulation shall meet the requirement specified in 3.2.24.1(b).

4.5.12 Parallel operation tests.- With each generator operating under the control of its control unit and with not more than one equalizer lead installed between control units as necessary, two generators and control units (and rectifier units if applicable) shall be operated in parallel for both increasing and decreasing loads at the speeds and loads specified herein. In addition, the generators shall be operated at the same speeds and at speeds so that the difference in speeds between the two generators is 400, 800 and 1200 r.p.m. and 1600 r.p.m. and with one generator

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operating at minimum full load speed and the other generator operating at maximum full load speed. Test results shall show conformance with 3.2.11.

4.5.13 Commutation.- Commutation shall be observed during all of the applicable tests specified herein, particularly at the maximum full load speed point. Commutation at all speeds and loads shall be black with no sparks showing or shall be with slight intermittent sparking.

4.5.14 Weight.- The weight of the d.c. generator and control unit or the a.c. generator, control unit and rectifier unit shall be taken and recorded on the test report.

4.5.15 Overspeed.- The generators shall be subjected to a speed 25 percent greater than the maximum full load speed for a period of 5 minutes. The generators shall be checked for noise, mechanical balance and smoothness of running during the test and for evidence of distortion, injury or noticeable change in the condition of any part after shut down.

4.5.16 Lubrication.- The performance of the lubricating system shall be carefully observed and inspected to determine whether it operates satisfactorily when the generator or alternator are operating in their normal horizontal position and during the inclined operation tests specified herein. The performance of the system shall be observed during the progress of other tests or by special tests as the circumstances may warrant.

4.5.17 Tests shall be conducted to determine that the cut-in speed, the load at which the load limiting device operates, the voltage at which the reverse current opens and closes and the voltage adjustment available in each voltage adjust device to show conformance with 3.2.24.1(a), 3.2.24.1(c) and 3.2.24.6. During this test, voltage regulator characteristics shall be observed and recorded for at least three points in the minimum full load to maximum full load speed range. These readings shall show conformance with 3.2.24.1(b).

4.5.18 Dynamic balance.- Generator armatures shall be balanced to show conformance with requirements of 3.2.17.13. The remaining unbalance and the type of machine used for balancing shall be recorded on the test report.

4.5.19 Battery chargers shall operate for 1000 hours under the following specified cyclic load conditions without replacing, adjusting or reconditioning of any part. The load shall consist of a battery of applicable voltage rating and a variable resistance.

(a) Fifteen minutes under an overload which will cause operation of the control unit load limiter when the generator is operating at minimum full load speed.
(b) Four hours at full load with the generator operating at minimum full load speed.
(c) Two hours at half load with the generator operating at minimum full load speed.
(d) Fifteen minutes with an overload which will cause operation of the control unit load limiter when the generator is operating at a speed corresponding to the one third speed point of the full load speed range. If the battery charger has a full load speed range of from 2000 to 5000 r.p.m., the one-third speed point of the full load speed range would correspond to a generator speed of 3,000 r.p.m.
(e) Three hours at full load with the generator operating at the speed specified in 4.5.19(d).
(f) Two hours at half load with the generator operating at the speed specified in 4.5.19(d).
(g) Fifteen minutes under an overload which will cause operation of the control unit load limiter when the generator is operating at a speed corresponding to the one half speed point of the full load speed range.
(h) Three hours at full load with the generator operating at the speed specified in 4.5.19(g).
(i) Two hours at half load with the generator operating at the speed specified in 4.5.19 (g).
(j) Fifteen minutes under an overload which will cause operation of the control unit load limiter when the generator is operating at a speed corresponding to the three-quarter speed point of the full load speed range.
(k) Three hours at full load with the generator operating at the speed specified in 4.5.19 (j).
(l) Two hours at half load with the generator operating at the speed specified in 4.5.19(j).
(m) One hour at full load with the generator operating at maximum full load speed.
(n) One hour at half load with the generator operating at maximum full load speed.
(o) One hour at no load with the generator operating at maximum full load speed.

4.5.19.1 During the 1000 hour test, the reverse current or load relay (if one is incorporated in the design) shall be operated so that it will disconnect the battery charger from the load at least 10,000 times.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The battery chargers, final technical manuals packed with battery chargers and repair parts, shall be preserved and packaged by level A or C; packed by level A, B or C and marked in accordance with MIL-E-17555 as specified (see 6.1). Preliminary and final drawings shall be packaged and packed as required by MIL-D-963.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2.1 and 1.2.2).
(c) Parallel operation if other than specified (see 3.2.11).
(d) Enclosure if other than specified (see 3.2.13).
(e) Special radio interference reduction requirements and requirements for filters and capacitors, (see 3.2.15.2 and 4.5.3).
(f) Generator dimensions, location of mounting holes, contour and speed-load characteristics and details of shaft extension (see 3.2.17.1 and 3.2.17.13).
(g) Details of belts, pulleys, mounting base and brackets, if required (see 3.2.17.1), and direction of rotation for Type I d.c. generators (see 3.2.17.2).
(h) Type of terminals and terminal box if other than specified (see 3.2.17.17).
(i) Engine mounting of control unit, if required (see 3.2.24.2).
(j) Special requirements for control unit terminal box, if other than specified (see 3.2.24.4).
(k) Weight, dimensions, contour characteristics and location of mounting holes for control unit (see 3.2.24.3).
(l) Number of sets of preliminary and final drawings. Types of drawings; whether drawing approval is required. (See 3.3.2, 3.3.2.1 and 3.3.2.2).
(m) Whether approval of manuals is required prior to printing. Where manual approval is required, specify number and distribution of preliminary manuals to be submitted for approval. Specify number and distribution of preliminary and final manuals (see 3.4).
(n) Requirements for onboard and stock repair parts (see 3.5).
(o) Requirements for preservation, packaging, packing and marking (see 5.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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APPENDIX I

SILICON RECTIFIER DIODES, SILICON VOLTAGE REFERENCE ELEMENTS,

VOLTAGE REGULATOR DEVICES AND TRANSISTORS.

10. SCOPE

10.1 This appendix lists a selection of silicon rectifier diodes, silicon voltage reference elements, regulator devices and transistors. This list is not an all-inclusive list and additions and deletions as applicable can be made by the manufacturer subject to the approval of the bureau or agency concerned.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-1/1108 - Semiconductor Device; Diode, Silicon Power Rectifier 1N1199-1206.

MIL-E-1/1134 - Zener Diode, Type 1N429.

MIL-E-1/1135 - Semiconductor Device; Diode, Silicon Power Rectifier 1N1183-1N1190.

MIL-E-1/1143 - Semiconductor Device; Silicon Junction Diode Types 1N645, 1N646, 1N647, 1N648, 1N649.

MIL-E-1/1202 - Semiconductor Devices, Diode; Silicon Power Rectifier Types 1N1396 through 1N1403.

MIL-T-19500/16 - Transistors, NPN, Silicon, Types JAN 2N342 and JAN 2N343.

MIL-S-19500/31 - Semiconductor Device, Transistor Type 2N341M.

MIL-T-19500/37 - Transistors, Type 2N332, 2N333, 2N334, and 2N335.

MIL-T-19500/54 - Transistor, Type 2N495.

MIL-T-19500/74 - Transistors, NPN, Silicon, Medium Power, Types JAN 2N497, JAN 2N498, JAN 2N656 and JAN 2N657.

MIL-T-19500/75 - Transistors, PN, Silicon Unijunction Types 2N489 - 2N494.

MIL-T-19500/78 - Transistors 2N1025, 2N1026 and 2N1026A.

MIL-T-19500/85 - Transistor, Type 2N496.

MIL-T-19500/99 - Transistors, Types 2N696 - 2N697.

MIL-S-19500/104 - Semiconductor Devices, Diode, Type 1N1124A, 1N1126A and 1N1128A.

MIL-T-19500/110 - Transistor, PNP, Silicon, Type 2N328A.

MIL-T-19500/111 - Transistor, PNP, Silicon, Type 2N329A.

MIL-S-19500/114 - Semiconductor Devices, Diode, 50-Watt, Silicon, B and RB Type, 1N2804 thru 1N2811, 1N2813, 1N2814, 1N2816, 1N2818 thru 1N2820, 1N2822 thru 1N2827, 1N2829, 1N2831 thru 1N2838 and 1N2840 thru 1N2846.

MIL-S-19500/115 - Semiconductor Devices, Diode, 1-Watt, Silicon Type 1N3016B thru 1N3051B.

MIL-S-19500/117 - Semiconductor Devices; Diodes, 400 mW, Types 1N962B through 1N992B.

MIL-S-19500/118 - Semiconductor Devices; Diodes, Silicon, Type 1N483B, 1N485B, 1N486B.

MIL-S-19500/124 - Semiconductor Diode, Silicon, 10-Watt, Voltage Regulator Types 1N2970B thru IN3015B and 1N2970RB thru 1N3015RB.

MIL-S-19500/127 - Semiconductor Devices, Diode, Types 1N746A thru 1N759A.

MIL-S-19500/162 - Semiconductor Diode, Power Rectifier, Types 1N614, 1N615, 1N616.

MIL-T-19500/173 - Transistor, Types 2N389 and 2N424.

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MILITARY (Cont'd.)

MIL-T-19500/176 - Transistors, Type 2N1047A, 2N1048A, 2N1049A, and 2N1050A.

MIL-T-19500/177 - Transistors, Types 2N1131 and 2N1132.

MIL-S-19500/194 - Semiconductor Devices, Diodes, Silicon, Power Rectifiers, Types JAN-1N253, JAN-1N254, JAN-1N255, and JAN-1N256.

MIL-S-19500/202 - Semiconductor Devices, Diodes, Silicon, Power Rectifiers, Types JAN-1N538M, JAN-1N540M, and JAN-1N547M.

MIL-S-19500/209 - Semiconductor Diodes, Silicon, Power Rectifier, Types 1N3164, 1N3168, 1N3170.

MIL-S-19500/240 - Semiconductor Devices, Diode, Silicon, Types JAN-1N645M, JAN-1N647M and JAN-1N649M.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. SILICON RECTIFIER DIODES.

30.1 Silicon rectifier diodes shall be selected in accordance with table III.

Table III - Silicon rectifier diodes.

JEDEC (EIA) No.	Maximum ratings up to temperature specified				Required derating at higher temp. MA/°C	Maximum oper. temp.		Detailed specification
	V_r 1/	I_F	T_A	T_C		T_A	T_C	
1N483B	70	125	75	-	1.0	150	-	MIL-S-19500/118
1N485B	180	125	75	-	1.0	150	-	MIL-S-19500/118
1N486B	225	125	75	-	1.0	150	-	MIL-S-19500/118
1N645M	225	250	75	-	2.5	150	-	MIL-S-19500/240
1N646	300	250	75	-	2.5	150	-	MIL-E-1/1143
1N647M	400	250	75	-	2.5	150	-	MIL-S-19500/240
1N648	500	250	75	-	2.5	150	-	MIL-E-1/1143
1N649M	600	250	75	-	2.5	150	-	MIL-S-19500/240
1N538M	200	500	85	-	4.0	150	-	MIL-S-19500/202
1N540M	400	500	85	-	4.0	150	-	MIL-S-19500/202
1N547M	600	500	85	-	4.0	150	-	MIL-S-19500/202
1N253	95	1.0A	-	135	73	-	150	MIL-S-19500/194
1N254	190	400	-	135	29	-	150	MIL-S-19500/194
1N255	380	400	-	135	29	-	150	MIL-S-19500/194
1N256	570	200	-	135	15	-	150	MIL-S-19500/194
1N1124AM	200	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1126AM	400	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1128AM	600	2.0A	-	100	20	-	150	MIL-S-19500/104
1N1614	200	4.5A	-	135	75	-	175	MIL-S-19500/162
1N1615	400	4.5A	-	135	75	-	175	MIL-S-19500/162
1N1616	600	4.5A	-	135	75	-	175	MIL-S-19500/162

See footnote at end of table.

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Table III - Silicon rectifier diodes (Cont'd).

JEDEC (EIA) No.	Maximum ratings up to temperature specified				Required derating at higher temp. MA/°C	Maximum oper. temp.		Detailed specification
	V_r 1/	I_F	T_A	T_C		T_A	T_C	
1N1199	50	12A	-	150	240	-	175	MIL-E-1/1108
1N1200	100	12A	-	150	240	-	175	MIL-E-1/1108
1N1201	150	12A	-	150	240	-	175	MIL-E-1/1108
1N1202	200	12A	-	150	240	-	175	MIL-E-1/1108
1N1203	300	12A	-	150	240	-	175	MIL-E-1/1108
1N1204	400	12A	-	150	240	-	175	MIL-E-1/1108
1N1205	500	12A	-	150	240	-	175	MIL-E-1/1108
1N1206	600	12A	-	150	240	-	175	MIL-E-1/1108
1N1183	50	35A	-	135	900	-	175	MIL-E-1/1135
1N1184	100	35A	-	135	900	-	175	MIL-E-1/1135
1N1185	150	35A	-	135	900	-	175	MIL-E-1/1135
1N1186	200	35A	-	135	900	-	175	MIL-E-1/1135
1N1187	300	35A	-	135	900	-	175	MIL-E-1/1135
1N1188	400	35A	-	135	900	-	175	MIL-E-1/1135
1N1189	500	35A	-	135	900	-	175	MIL-E-1/1135
1N1190	600	35A	-	135	900	-	175	MIL-E-1/1135
1N2172	50	50A	-	150	2000	-	175	MIL-S-19500/209
1N2173	100	50A	-	150	2000	-	175	MIL-S-19500/209
1N2174	200	50A	-	150	2000	-	175	MIL-S-19500/209
1N1682	300	50A	-	150	2000	-	175	MIL-S-19500/209
1N1396	50	70A	-	150	2800	-	175	MIL-E-1/1202
1N1397	100	70A	-	150	2800	-	175	MIL-E-1/1202
1N1398	150	70A	-	150	2800	-	175	MIL-E-1/1202
1N1399	200	70A	-	150	2800	-	175	MIL-E-1/1202
1N1400	300	70A	-	150	2800	-	175	MIL-E-1/1202
1N1401	400	70A	-	150	2800	-	175	MIL-E-1/1202
1N1402	500	70A	-	150	2800	-	175	MIL-E-1/1202
1N1403	600	70A	-	150	2800	-	175	MIL-E-1/1202

1/ RMS ac voltage shall not exceed $0.7V_r$.

Symbols: V_r - Peak reverse voltage across diode, dc volts.

I_F - Average current in forward direction, dc milliamperes (except dc amperes where "A" appears).

T_A - Ambient temperature at diode, degrees centigrade.

T_C - Temperature on or adjacent to mounting surface of the diode, degrees centigrade.

40. SILICON VOLTAGE REFERENCE ELEMENTS

40.1 Silicon voltage reference elements shall be selected in accordance with Table IV.

50. VOLTAGE REGULATOR DEVICES

50.1 Voltage regulator devices shall be selected in accordance with table V.

60. TRANSISTORS

60.1 Transistors shall be selected in accordance with table VI.

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Table IV - Silicon voltage reference elements.

JEDEC (EIA) no.	Design values			Maximum ratings		Detailed specification
	Reference voltage V_{ZR} @ I_{ZB} DC VOLTS		-55°C to 100°C Temperature error ΔV_{ZRT} @ I_{ZB} DC MV	Bias current range[1/] I_{ZB} DC MA	Ambient temperature at reference element	
	min	max				
1N429	5.9	6.5	± 50	7.5 ± 2%	-55 to 100°C	MIL-E-1/1134

[1/] Temperature error ΔV_{ZRT} is not valid for reverse current outside this range.

Symbols: V_{ZR} - Reverse voltage at a specified reverse bias current, I_{ZB}.

ΔV_{ZRT} - Maximum error in V_{ZR} over the specified temperature range at I_{ZB}.

I_{ZB} - Reverse current to which element is biased to obtain design value performance.

Table V - Voltage regulator devices.

Power dissipation (nominal 25°C value)	Detailed specification
400 mw (3.3 to 10 volts)	MIL-S-19500/127
400 mw (11 to 200 volts)	MIL-S-19500/117
1 watt (6.8 to 200 volts)	MIL-S-19500/115
10 watts (6.8 to 200 volts)	MIL-S-19500/124
50 watts (6.8 to 200 volts)	MIL-S-19500/114

Note: Derate in accordance with the applicable detailed specification indicated above.

Table VI - Transistors.

JEDEC (EIA) No.	Type	Gain h_{fe}	Maximum ratings							Detailed specification
			V_{cb} dc Volts	I_c Avg ma	P_c up to temperature specified		Required derating at higher temp. mw/°C	Maximum oper. temp.		
					T_A 65°C	T_C 100°C		T_A	T_C	
2N328AM	PNP	18-44	50	50	250	-	2.9	150	-	MIL-T-19500/110
2N329AM	PNP	36-88	50	50	250	-	2.9	150	-	MIL-T-19500/111
2N333	NPN	18-40	45	25	100	-	1.2	150	-	MIL-T-19500/37
2N335	NPN	38-91	45	25	100	-	1.2	150	-	MIL-T-19500/37
2N337	NPN	5 MIN	45	20	85	-	1.0	150	-	MIL-T-19500/69
2N338	NPN	10 MIN	45	20	85	-	1.0	150	-	MIL-T-19500/69

See footnotes at end of table.

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Table VI - Transistors (Cont'd).

JEDEC (EIA) No.	Type	Gain h_{fe}	Maximum ratings							Detailed specification
			V_{cb} dc Volts	I_c Avg ma	P_c up to temperature specified		Required derating at higher temp. mw/°C	Maximum oper. temp.		
					T_A 65°C	T_C 100°C		T_A	T_C	
2N341M	NPN	9-90	125	40	680	-	8.0	150	-	MIL-S-19500/31
2N343	NPN	28-90	1/ 60	60	680	-	8.0	150	-	MIL-T-19500/16
2N389	NPN	12-60	1/ 60	1500	-	48.5w	485	-	150	MIL-S-19500/173
2N424	NPN	12-60	1/ 80	750	-	48.5w	485	-	150	MIL-S-19500/173
2N491	2/	-	3/ 60	4/ 70	300	-	3.6	150	-	MIL-T-19500/75
2N495	PNP	9 MIN	25	50	100	-	1.3	140	-	MIL-T-19500/54
2N496	PNP	1.8 MIN	10	50	100	-	1.3	140	-	MIL-T-19500/85
2N498	NPN	12-36	100	200	-	2.2w	24	-	150	MIL-T-19500/74
2N657	NPN	30-90	100	200	-	2.2w	24	-	150	MIL-T-19500/74
2N697M	NPN	40-120	1/ 40	150	-	950	14	-	150	MIL-S-19500/99
2N1026	PNP	18-36	1/ 35	100	100	-	1.2	150	-	MIL-S-19500/78
2N1050	NPN	30-90	1/ 120	500	-	22w	240	-	150	MIL-S-19500/176
2N1132	PNP	2.5 AVG	40	600	-	950	1.4	-	150	MIL-T-19500/177

1/ V_{ce} - Collector-to-emitter voltage, dc volts.

2/ - Device has one junction (emitter) contact and two ohmic (base) contacts on n-type crystal.

3/ V_{B2E} - Base two-to-emitter voltage, dc volts.

4/ I_e - Emitter current, rms milliamperes.

Symbols: h_{fe} - Small signal short-circuit forward-transfer current ratio (common emitter base input) - equals Beta.
P_c - Collector power dissipation, milliwatts average (watts average where "w" appears).
V_{cb} - Collector voltage, I_c - Collector current.
T_A - Ambient temperature at transistor, T_C - Transistor case or body temperature on or adjacent to thermal conduction mounting surface.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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